# Top 50 Statistics Questions Every Data Analyst Must Know (Interview Ready 🚀) Connect With me 1:1

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**Part 1: Descriptive Statistics & Foundations**

1. **What is the difference between Descriptive and Inferential statistics?**

Descriptive statistics summarize data (mean, median, mode), while Inferential statistics use a sample to make predictions or generalizations about a larger population.

2. **Define the Central Limit Theorem (CLT).**

The CLT states that the sampling distribution of the mean will approach a normal distribution as the sample size increases, regardless of the population's original distribution shape (usually $n \geq 30$).

3. **What is a Normal Distribution?**

A bell-shaped symmetric distribution where the mean, median, and mode are all equal. Approximately 68% of data falls within one standard deviation, 95% within two, and 99.7% within three.

4. **What is the difference between Mean and Median?**

The mean is the average (sensitive to outliers), while the median is the middle value (robust against outliers).

5. **When should you use the Median over the Mean?**

When the data is skewed or contains significant outliers (e.g., household income).

6. **What is Standard Deviation?**

A measure of the amount of variation or dispersion in a set of values. A low SD indicates data points are close to the mean.

7. **What is Variance?**

The average of the squared differences from the Mean. It measures how far a set of numbers are spread out from their average value.

8. **Define "Skewness."**

Skewness measures the asymmetry of a distribution. Positive skew means the tail is on the right; negative skew means the tail is on the left.

9. **What is Kurtosis?**

Kurtosis measures the "tailedness" of a distribution. High kurtosis indicates more extreme outliers (heavy tails) than a normal distribution.

10. **Explain the Interquartile Range (IQR).**

The difference between the third quartile (Q3) and the first quartile (Q1). It represents the middle 50% of the data.

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**Part 2: Probability & Sampling**

11. **What is the difference between "Probability" and "Likelihood"?**

Probability quantifies outcomes given a fixed parameter. Likelihood is used to estimate unknown parameters based on observed data.

12. **What is a P-value?**

The probability of obtaining test results at least as extreme as the results actually observed, under the assumption that the null hypothesis is correct.

13. **Define Type I and Type II Errors.**

   - **Type I:** Rejecting a true null hypothesis ("False Positive").
   - **Type II:** Failing to reject a false null hypothesis ("False Negative").

14. **What is a Confidence Interval?**

A range of values, derived from sample statistics, that is likely to contain the value of an unknown population parameter.

15. **Explain the Law of Large Numbers.**

As a sample size grows, its mean gets closer to the average of the whole population.

16. **What is Sampling Bias?**

When the sample collected does not accurately represent the population, leading to skewed results.

17. **What is Simple Random Sampling?**

Every member of the population has an equal chance of being selected.

18. **What is Stratified Sampling?**

Dividing the population into subgroups (strata) and then randomly sampling from each stratum to ensure representation.

19. **What is Cluster Sampling?**

Dividing the population into clusters (usually by geography) and then randomly selecting entire clusters to study.

20. **Define the "Margin of Error."**

The amount of random sampling error in a survey's results.

## Part 3: Hypothesis Testing

21. **What are the steps of Hypothesis Testing?**

State Null ($H_0$) and Alternative ($H_1$) hypotheses, choose a significance level ($\alpha$), calculate the test statistic, determine the p-value, and make a decision.

22. **What is the Null Hypothesis?**

The assumption that there is no effect or no difference.

23. **What is an Alternative Hypothesis?**

The hypothesis that there is a significant effect or difference.

24. **What is a Z-test vs. a T-test?**

Use a **Z-test** when the population variance is known and sample size is large. Use a **T-test** when the population variance is unknown and sample size is small.

25. **What is a One-tailed vs. Two-tailed test?**

A **One-tailed** test looks for a change in one direction. A **Two-tailed** test looks for any change (increase or decrease).

26. **Explain ANOVA.**

Analysis of Variance (ANOVA) is used to compare the means of three or more independent groups to see if at least one is significantly different.

27. **What is the Chi-square test?**

A statistical test used to determine if there is a significant association between two categorical variables.

28. **What is Power in statistics?**

The probability that a test will correctly reject a false null hypothesis ($1 - \beta$).

29. **What is a Significance Level ($\alpha$)?**

The threshold for rejecting the null hypothesis, commonly set at 0.05.

30. **What is the degrees of freedom?**

The number of independent values or quantities which can be assigned to a statistical distribution.

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## Part 4: Correlation & Regression

31. **What is Correlation?**

A measure of the linear relationship between two variables, ranging from -1 to +1.

32. **Does Correlation imply Causation?**

No. Just because two variables move together doesn't mean one causes the other (could be a third "lurking" variable).

33. **What is Linear Regression?**

A method to model the relationship between a dependent variable and one or more independent variables.

34. **What are the assumptions of Linear Regression?**

Linearity, Independence, Homoscedasticity, and Normality of errors.

35. **What is $R$-squared?**

A statistical measure that represents the proportion of the variance for a dependent variable that's explained by an independent variable.

36. **What is Multicollinearity?**

When independent variables in a regression model are highly correlated with each other, making it hard to determine their individual effects.

37. **What is Homoscedasticity?**

A condition where the variance of the residual terms is constant across all levels of the independent variables.

38. **Define "Overfitting."**

When a model is too complex and follows the noise in the data rather than the underlying pattern.

39. **Define "Underfitting."**

When a model is too simple to capture the underlying trend of the data.

40. **What is Logistic Regression?**

Used when the dependent variable is categorical (e.g., Yes/No, 0/1).

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**Part 5: Advanced Concepts & Data Handling**

41. **What is a Box Plot?**

A standardized way of displaying the distribution of data based on a five-number summary: minimum, Q1, median, Q3, and maximum.

42. **How do you handle outliers?**

Depending on the context: remove them, transform the data (log), or use robust statistical methods.

43. **What is Simpson’s Paradox?**

A phenomenon where a trend appears in several different groups of data but disappears or reverses when these groups are combined.

44. **What is Bayes' Theorem?**

A formula that describes the probability of an event based on prior knowledge of conditions that might be related to the event.

45. **What is the difference between Point Estimate and Interval Estimate?**

A point estimate is a single value (e.g., mean), while an interval estimate provides a range of values (e.g., confidence interval).

46. **What is A/B Testing?**

A statistical way of comparing two versions of a variable (e.g., webpage A vs. B) to determine which performs better.

47. **Define "Standard Error."**

The standard deviation of the sampling distribution of a statistic (most commonly the mean).

48. **What is the Poisson Distribution?**

A distribution that models the number of times an event occurs in a fixed interval of time or space.

49. **What is Cross-Validation?**

A technique for assessing how the results of a statistical analysis will generalize to an independent data set.

50. **What is the p-hacking?**

The misuse of data analysis to find patterns which can be presented as statistically significant when they are actually not.

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